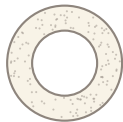


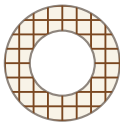
a What each acquisition supplies

Routine cine



boundaries +
weak texture

Tagging



prepared grid
(material pattern)

DENSE



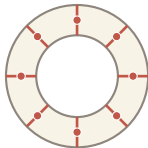
phase-encoded
displacement

Only tagging and DENSE carry a material label;
cine supplies boundaries and weak texture.

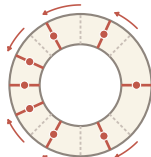
b Two mappings, identical contours

Mapping 1: $\theta \mapsto \theta$

Mapping 2: $\theta \mapsto \theta + a \sin \theta$



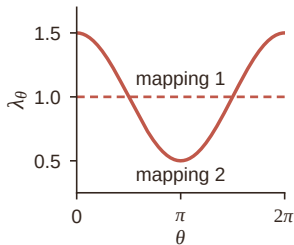
identical
masks



Dice = 1, Hausdorff = 0; $r \mapsto r$, $a = 0.5$

Dashed: reference positions. Solid: mapped material
points (same eight points in both panels).

c Resulting stretch



$$\lambda_{\theta} = 1 + a \cos \theta$$

range 0.5–1.5; orientation
preserved for $|a| < 1$

Analytic example; no cardiac measurement is shown. The circumferential stretch ratio is a different quantity from Green–Lagrange strain.